

PORTFOLIO

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Nuryabilla Utami, S.T.

Fresh Graduate 2026



Geodetic Engineering

Universitas Diponegoro

INTRODUCTION



 Cirebon, Indonesia

EDUCATION

Universitas Diponegoro 2022-2026

S1 - Geodetic Engineering
CGPA 3.57/4.00



SCAN CV HERE

ABOUT ME

Fresh graduate Bachelor's of Geodetic Engineering Universitas Diponegoro (2026) with internship experience in **surveying, mapping, GIS, and remote sensing**. Proficient in spatial data processing using **ArcGIS, QGIS, AutoCAD, Google Earth Engine, ENVI, SNAP, and Agisoft Metashape**, as well as data management and automation using **Python (Geospatial), SQL, and PostgreSQL**. Experienced in operating field surveying instruments such as **Total Station, Waterpass (Automatic Level), and GNSS**. Detail oriented and ready to contribute professionally in the engineer and geoscience field.

SOCIAL MEDIA AND CONTACT

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 github.com/nuryuwii-cue

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Curriculum Vitae (CV)

 visit me on:
Website Portfolio

MY EXPERTISE ON

GEODETTIC ENGINEERING

CORE EXPERTISE

SURVEYING & MAPPING

Utilizing ground surveying instruments (Total Station, Waterpass, GNSS) and CAD software (AutoCAD/Civil 3D) to perform **stakeout**, **generate contours**, **create cross & long sections**, **calculate cut & fill volumes**, and **deliver accurate as-built drawings**.

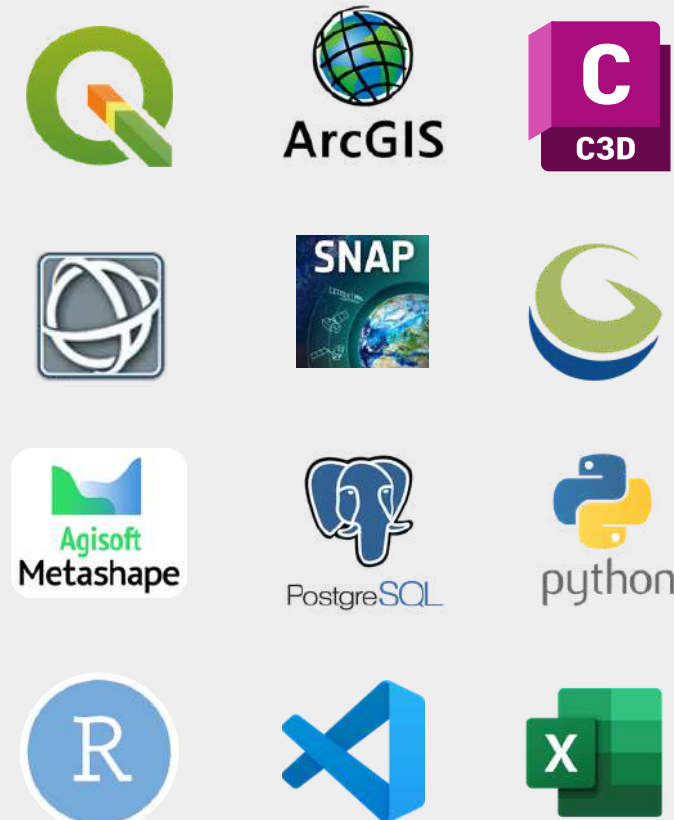
GIS & REMOTE SENSING

Utilizing ArcGIS, QGIS, SQL, ENVI, and Python to process spatial data, integrate active & passive remote sensing data processing (**SAR, NDVI, Landuse**) and **create thematic maps** to generate insights and support machine learning models.

PHOTOGRAMMETRY

Planning UAV flight paths and processing aerial & close-range photogrammetry data using Agisoft Metashape, Global Mapper, and Trimble software to generate **high-resolution 3D models**, **Digital Elevation Models (DEM)**, and **Digital Surface Models (DSM)**.

TOOLS & SOFTWARE



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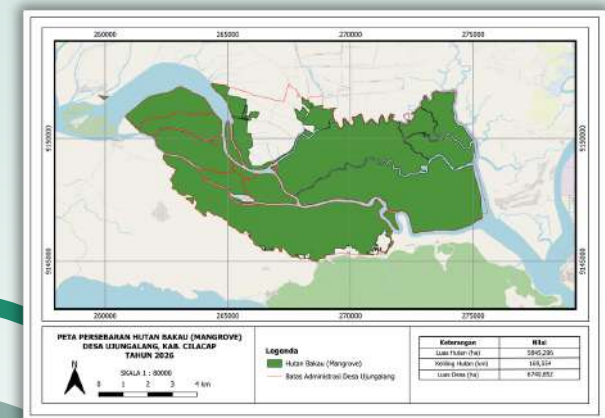
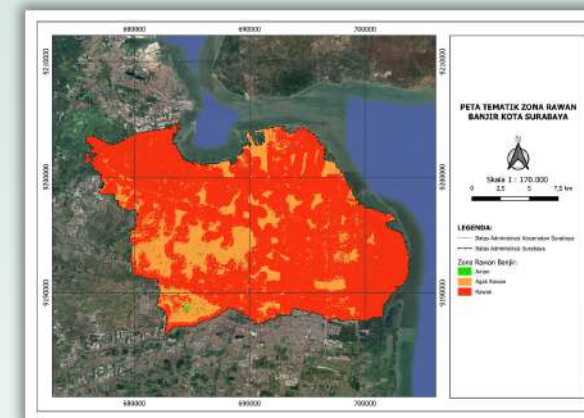
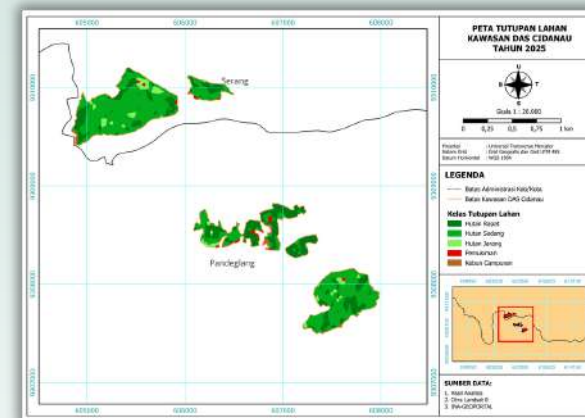
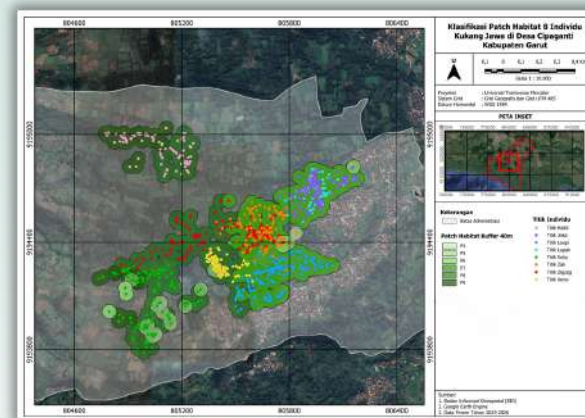
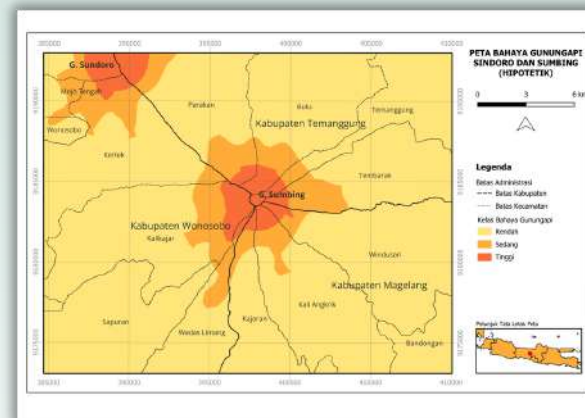
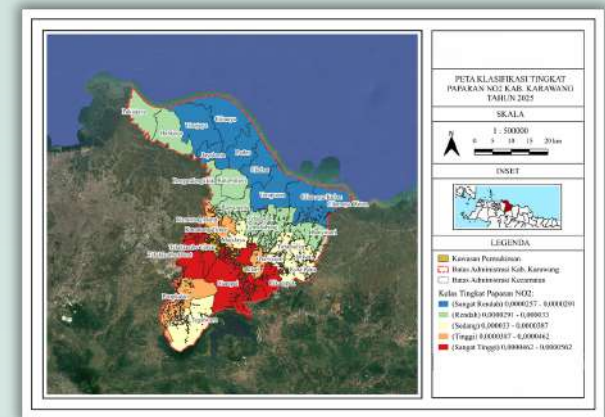
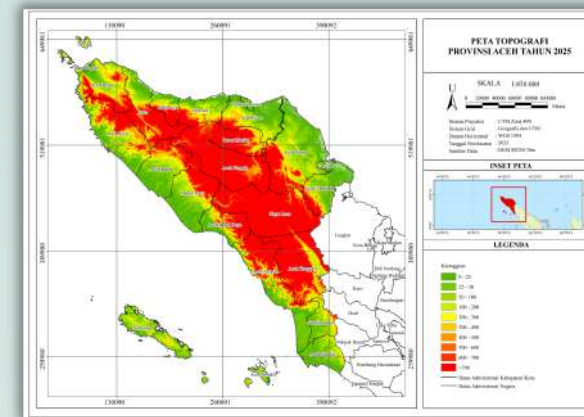
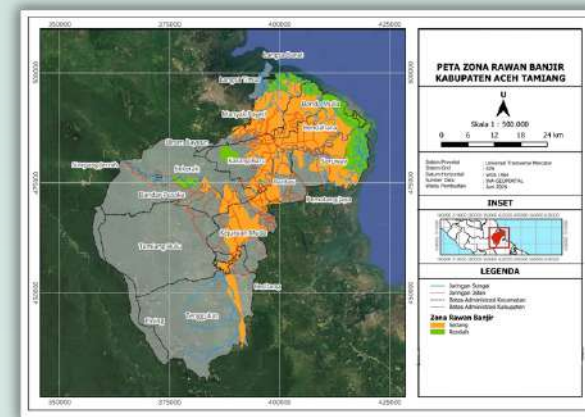
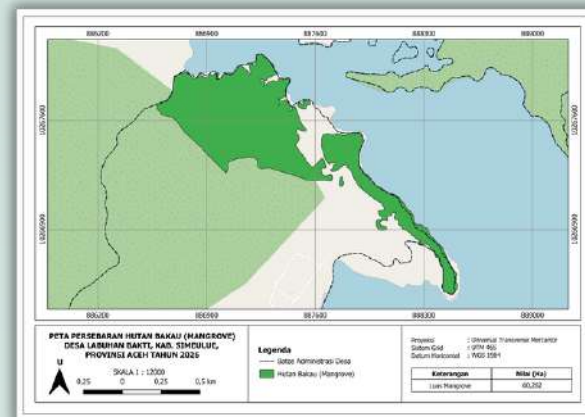
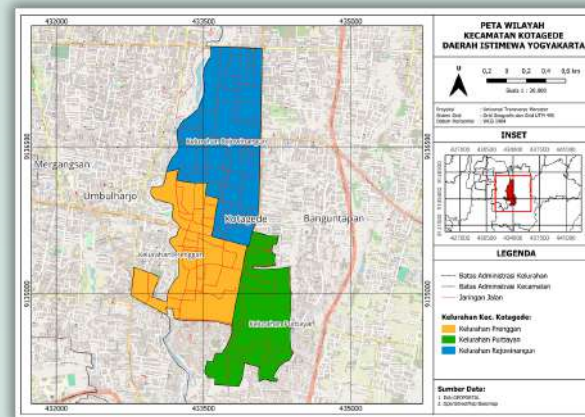
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PROFESSIONAL EXPERIENCES

GIS SPECIALIST - FREELANCE

May 2026 - Present

- Completed 25+ geospatial projects, including thematic map production and technical consulting in geospatial fields for clients from academic and private sectors.



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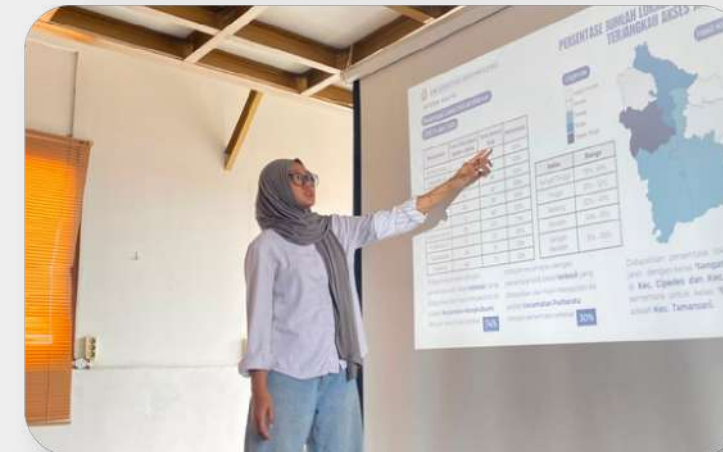
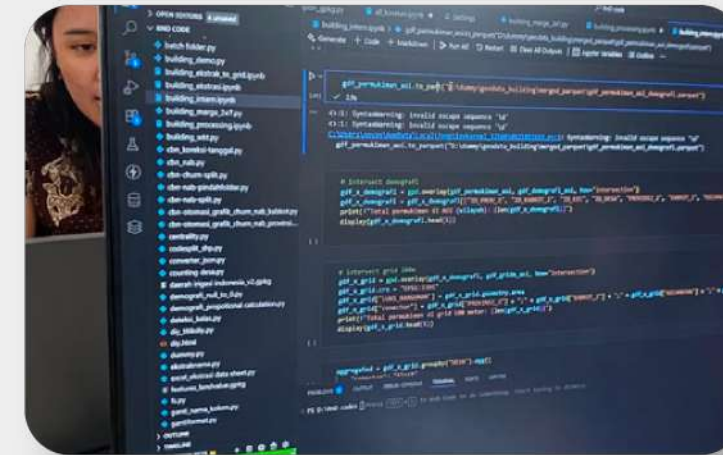
INTERNSHIP EXPERIENCES

GIS ANALYST INTERN

PT MULTI AREAL PLANING INDONESIA (MAPID)

June - August 2025

- **Surveys potential POIs across 9 cities (Kedungsepur & Sarbagita) to update spatial catalog data.**
- **Processed building footprint downscaling (25 cities) and LST data using Python.**
- **Completed an individual project titled "Spatial Analysis of Potential Employment Absorption Areas by District in Tasikmalaya City, 2024" using QGIS.**



SURVEYOR INTERN

ATR BPN KOTA CIREBON

January - February 2025

- **Land parcel surveys and prepared survey drawings/plans using AutoCAD for land certificate documentation.**
- **Performed verification and examination of land certificate documents and legal records for 22 sub-district in Cirebon.**



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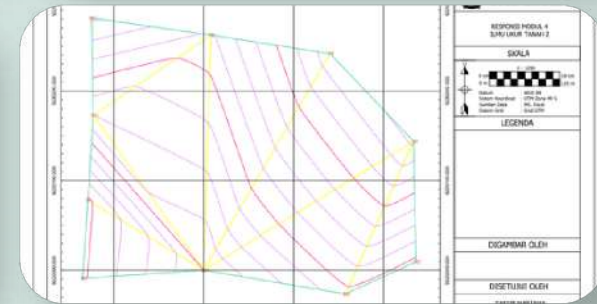
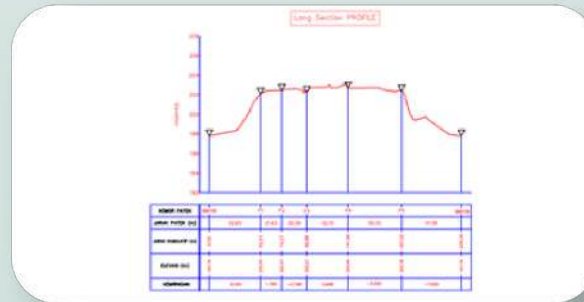
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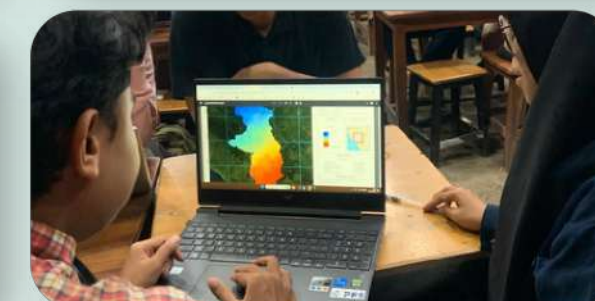
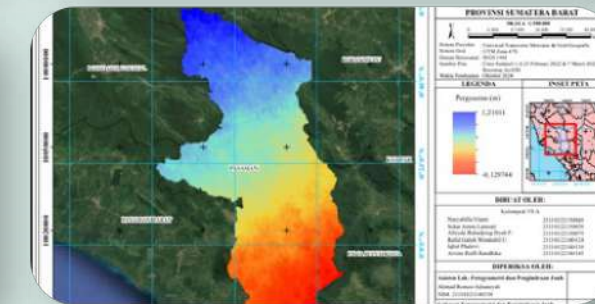
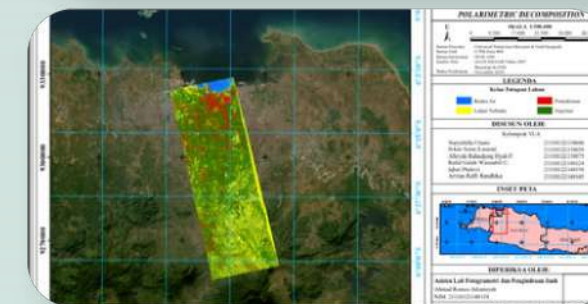
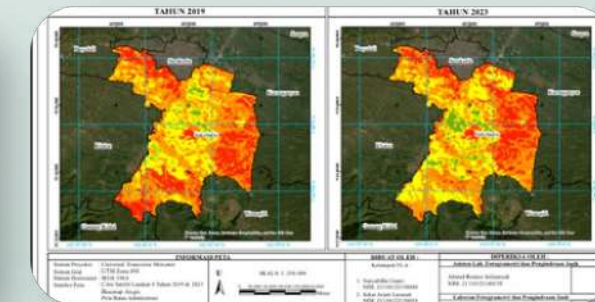
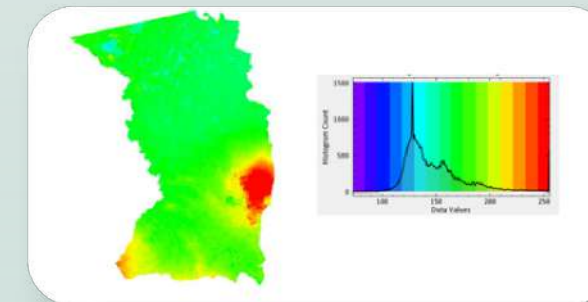
ACADEMY EXPERIENCES

PRACTICUM: TOPOGRAPHIC SURVEYING



- Executed terrestrial and GNSS surveys using Total Station, Theodolite, Waterpass, leveling rod, and measuring tape for contour, long, and cross-section mapping.
- Calculated cut & fill volumes, performed stake-outs, and drafted as-built drawings using AutoCAD Civil 3D.

PRACTICUM: REMOTE SENSING



- Mapped land use from Landsat 8 using supervised and unsupervised machine learning methods, and extracted spectral indices (NDVI, NDBI, NDWI, LST).
- Processed Sentinel-1 SAR data using DInSAR for deformation analysis and land use using Polarimetric Decomposition techniques.



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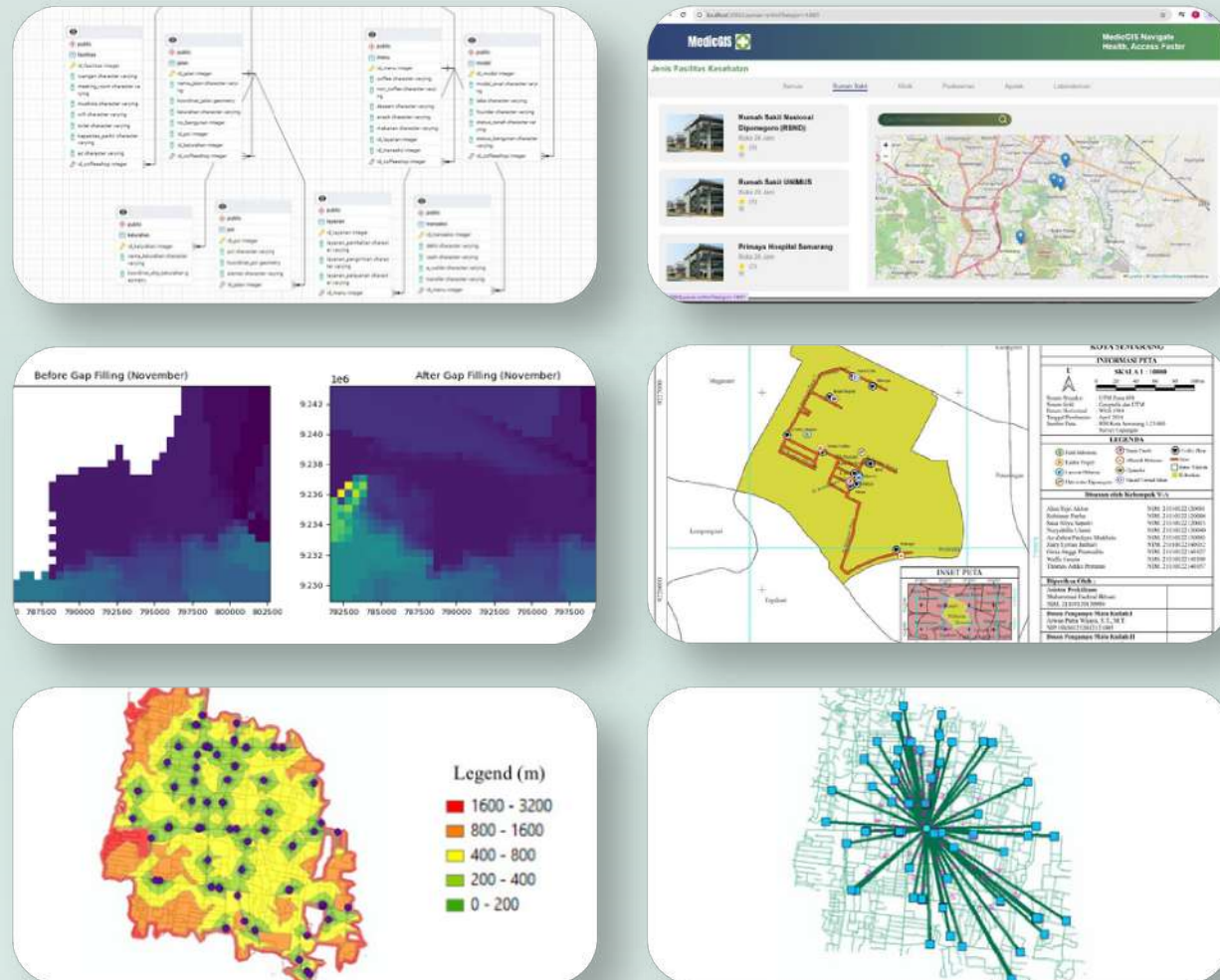
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ACADEMY EXPERIENCES

PRACTICUM: DBMS & GIS



- Designed ERD models for PostgreSQL database management and developed WebGIS applications for Health Facilities in Tembalang.
- Applied Machine Learning modeling for geospatial data, create Service Area, OD Cost Matrix, and Closest Facility analysis.

PRACTICUM: PHOTOGRAMMETRY



- Designed flight paths for aerial photo acquisition and generated stereo photo pairs.
- Created ortho-mosaics and DEM/DSM/DBM elevation models.
- Reconstructed 3D object models using Close Range Photogrammetry techniques.



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ACADEMY PROJECT

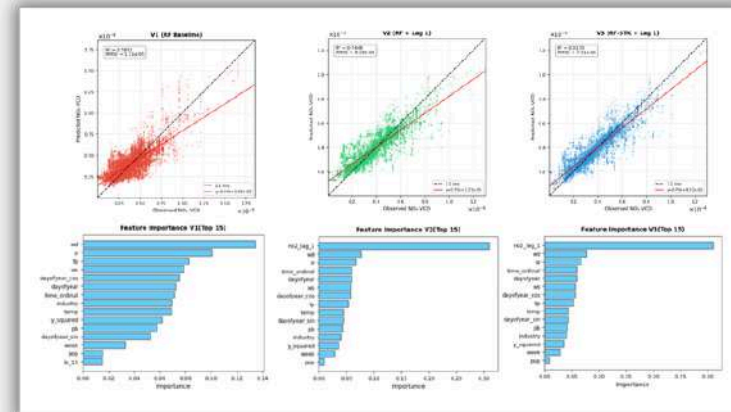
FINAL PROJECT RESEARCH (2026)

Geostatistical Modeling of NO₂ Concentrations and Its Impact on Acute Respiratory Infections (ARI) Prevalence in Mid-2025 in Semarang City

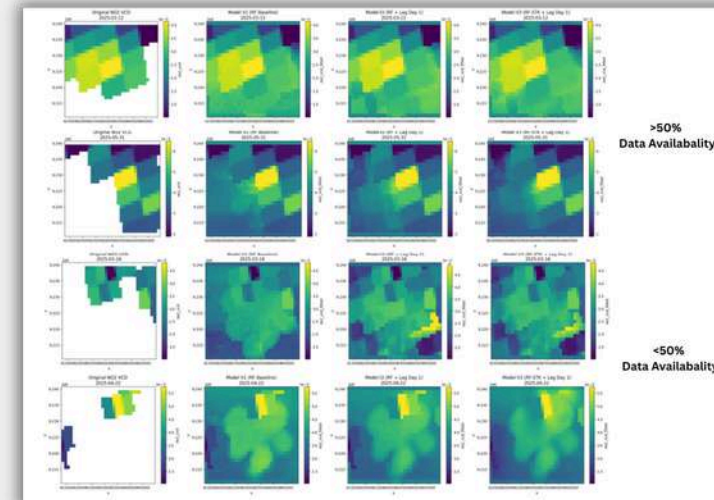
BACKGROUND OF RESEARCH

- Indonesia drives 75% of global air pollution burden; Semarang recorded AQI 147 and 154k+ ARI cases in mid-2025.
- Assesses carcinogenic NO₂ concentration influence on high local ARI prevalence.
- Employs daily Sentinel-5P satellite monitoring while addressing data gaps caused by cloud cover and QA filter limits (≥ 0.75) using machine learning model (RF0, RFLagDay1, RFSTK).

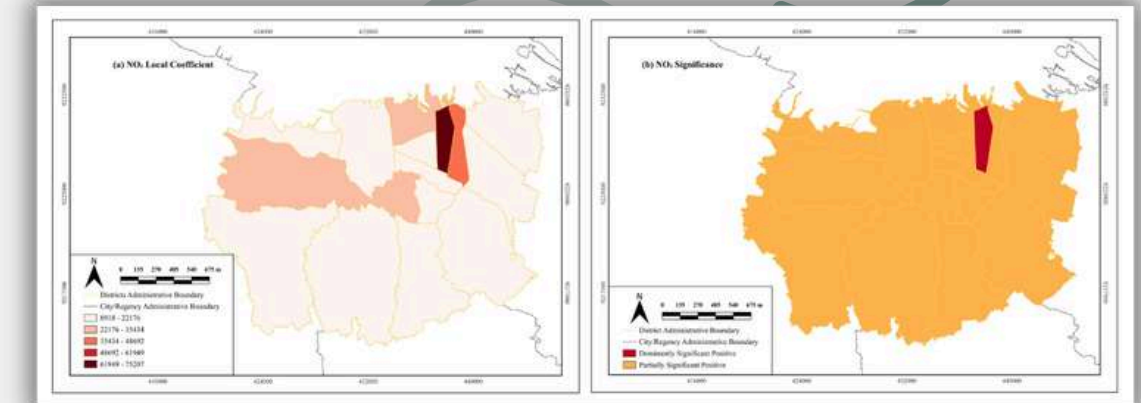
RESULTS & ANALYSIS



RF-STK (V3) yielded **highest accuracy ($R^2 = 0.80$)**, driven by NO₂ lag-1, wind direction, and solar radiation.



- **>50% Data:** V3 achieved best spatial coherence via Kriging, outperforming V1 (coarse) and V2 (lag-1 feature).
- **<50% Data:** Reliability dropped for all models as predictions relied more on predictors than satellite data.



- GTWR superior to OLS (R^2 : **0.681**; Moran's I: **0.467**).
- Universal **positive** NO₂–ARI impact, strongest in **East Semarang** (traffic/industry) and peaking in **dry season (Jan & Feb–Mar)**.



OPEN PROJECT



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ACADEMY PROJECT

KKN INDIVIDUAL PROJECT (2025)

PROJECT 1

Distribution Map of Avocado Ketapang Tree Varieties of BUMDes.



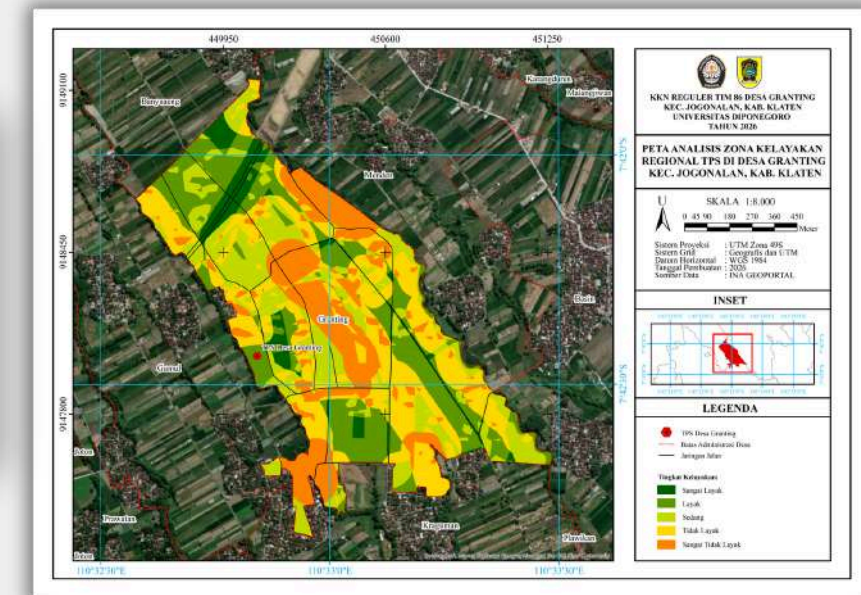
This avocado variety distribution map, derived from the spatial database, serves as a data-driven tool for BUMDes managers to plan harvests, monitor tree health, and develop targeted marketing and orchard management strategies.



PROJECT 2

Development of a Regional Feasibility Zone Analysis Map for Waste Disposal Sites (TPS) in Granting Village, Jogonalan District, Klaten Regency.

This map helps Granting Village make data-driven decisions to select and manage safe, sustainable TPS 3R sites. It supports environmental and social risk mitigation, improves spatial planning, and guides future village waste management and development programs.



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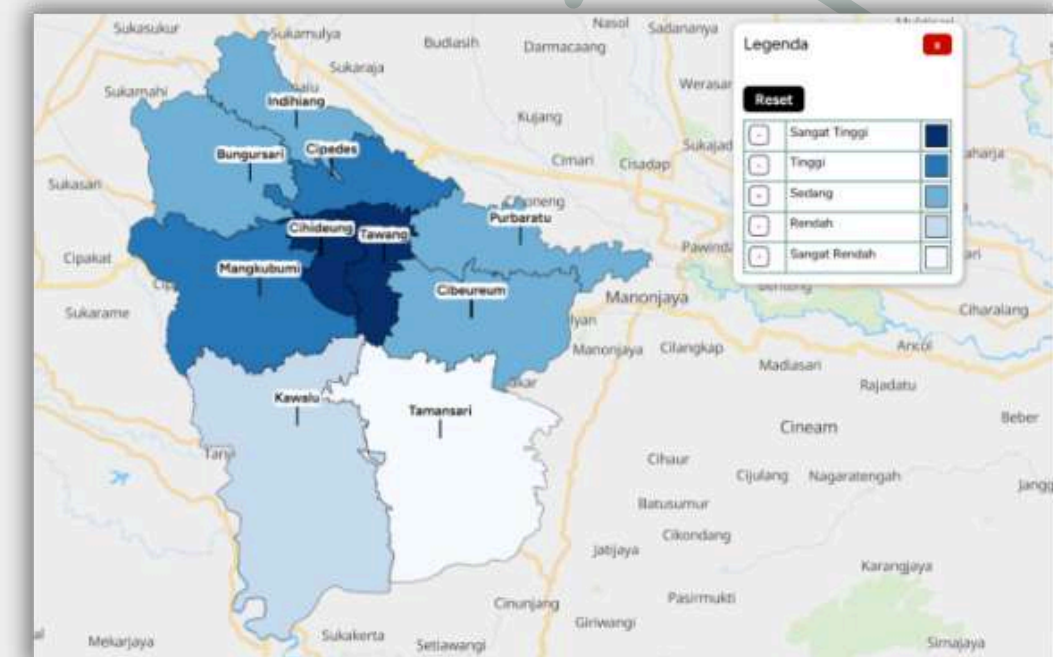
ACADEMY PROJECT

INTERNSHIP INDIVIDUAL PROJECT (2025)

Spatial Analysis for Mapping Potential Labor Absorption Areas by Sub-District using Criteria Importance Through Intercriteria Correlation (CRITIC) Method in Tasikmalaya, Jawa Barat, 2024

BACKGROUND OF RESEARCH

- Tasikmalaya's economy (10 sub-districts) is driven mainly by trade, construction, and manufacturing.
- A high participation rate (68.92%) contrasts sharply with job seekers (2,619) outnumbering available roles (1,067).
- Unequal job distribution across sub-districts worsens regional inequality and fuels unemployment.



Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	387 - 394	5		0.289
Tinggi	344 - 356	4		0.176
Sedang	299 - 311	3	0.28	0.847
Rendah	211 - 281	2		0.226
Sangat Rendah	185 - 282	1		0.145

(a)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	13.41 - 11.136	5		0.177
Tinggi	11.136 - 4.873	4		0.169
Sedang	4.873 - 2.354	3	0.18	0.442
Rendah	2.354 - 1.972	2		0.295
Sangat Rendah	1.972 - 1.888	1		0.147

(b)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	9138 - 8324	5		0.855
Tinggi	8109 - 2941	4		0.642
Sedang	3945 - 2543	3	0.28	0.482
Rendah	2548 - 1736	2		0.321
Sangat Rendah	1708 - 1579	1		0.181

(c)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	287 - 212	5		0.823
Tinggi	212 - 79	4		0.666
Sedang	79 - 43	3	0.17	0.51
Rendah	43 - 22	2		0.333
Sangat Rendah	22 - 21	1		0.167

(d)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	0.73 - 0.88	5		0.609
Tinggi	0.88 - 0.53	4		0.487
Sedang	0.53 - 0.32	3	0.12	0.369
Rendah	0.32 - 0.31	2		0.244
Sangat Rendah	0.31 - 0.3	1		0.122

(e)

Kelas	Range	Skor	Bobot	Skor x Bobot
Sangat Tinggi	12324 - 28792	5		0.574
Tinggi	28792 - 25076	4		0.439
Sedang	25076 - 18392	3	0.11	0.344
Rendah	18392 - 12854	2		0.25
Sangat Rendah	12854 - 12854	1		0.115

(f)

Labor absorption in Tasikmalaya is concentrated in the **city center** due to **strong infrastructure**, while **peripheral areas lag behind**.

(a) Job Seekers; (b) Industrial Density; (c) Productive Age Pop;
(d) Job Vacancies; (e) Industrial Road Access; (f) Higher Education Pop

[OPEN PROJECT](#)

THANK YOU

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